



CPSC 100

Computational Thinking

Intro to Programming

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Agenda

- Learning Goals
- Course Admin
- Intro to Programming [Continued]
 - Boolean Functions/Expressions
 - Repeat Blocks → For Loops

Course Admin



Learning Goals

After this week's lecture, you should be able to:

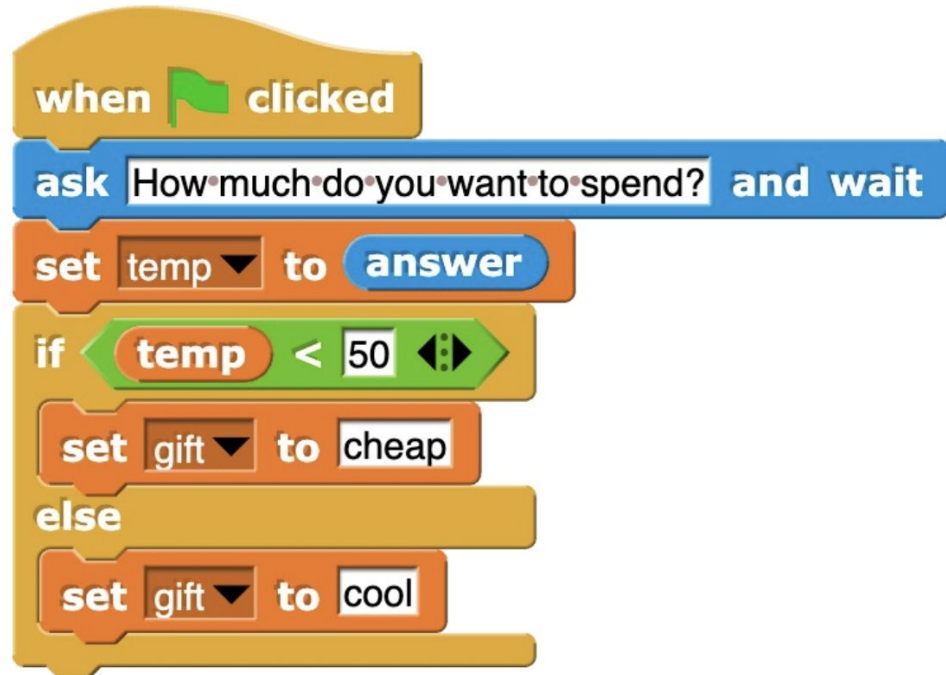
- Define **boolean expressions** and their role in Snap! Programming
- **Apply** boolean operations to control flow in programming
- Describe the concept of **loops** and **iteration** in programming
 - Differentiate between **finite loops** (repeat N times) and **condition-based loops** (repeat until)
- Use logical reasoning to **determine the output of given code**
 - Apply CT to **trace and evaluate** code snippets

Q: What is the value of gift after the block is run, assume user input is 100?



iClicker

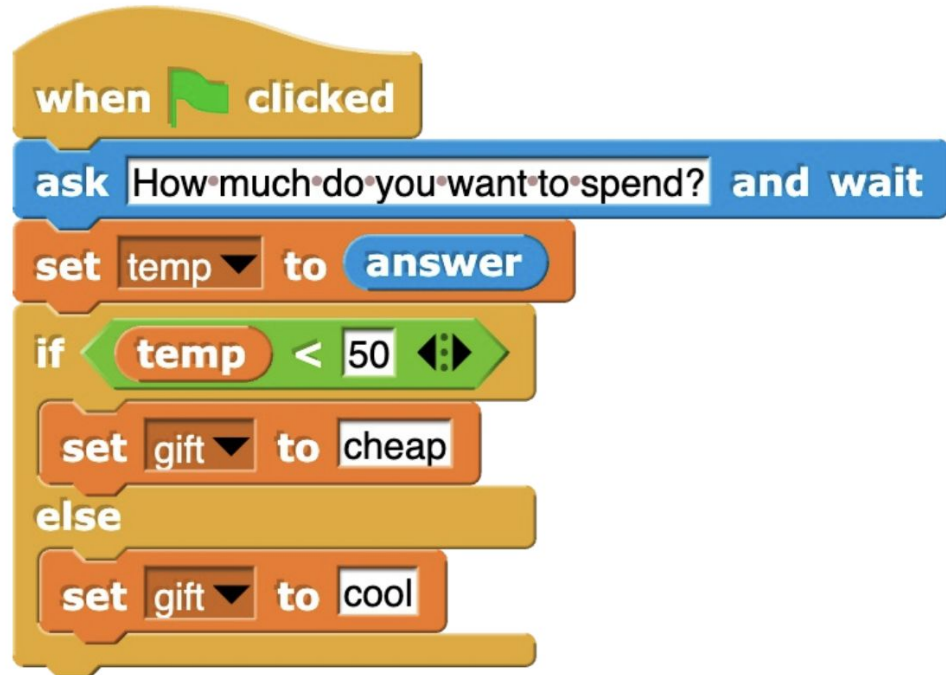
- A. 50
- B. 100
- C. 150
- D. cheap
- E. cool



Q: What is the value of gift after the block is run, assume user input is 50?



- A. 50
- B. 100
- C. 150
- D. cheap
- E. cool



Slides for Pre-reading

Programming Basics

Programming Basics

0. Variable



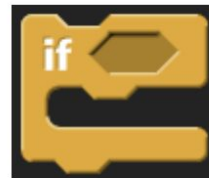
- a. A named storage location for data

1. Sequencing

- a. In order (top-down)

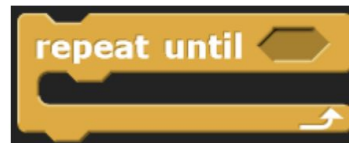
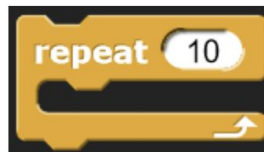
2. Selection

- a. Conditional Structures



3. Iteration

- a. Loop Structures



Components of an Algorithm

0. Variable

- A named storage location for data



1. Sequencing

- In order (top-down)

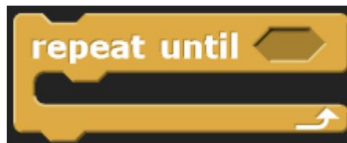
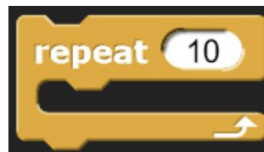
2. Selection

- Conditional Structures



3. Iteration

- Loop Structures



Boolean (or Logical) Function





Boolean (or Logical) Function

At the very lowest level, computer circuitry is made of wires, and each wire either has power or it does not: meaning it's either **on or off**.

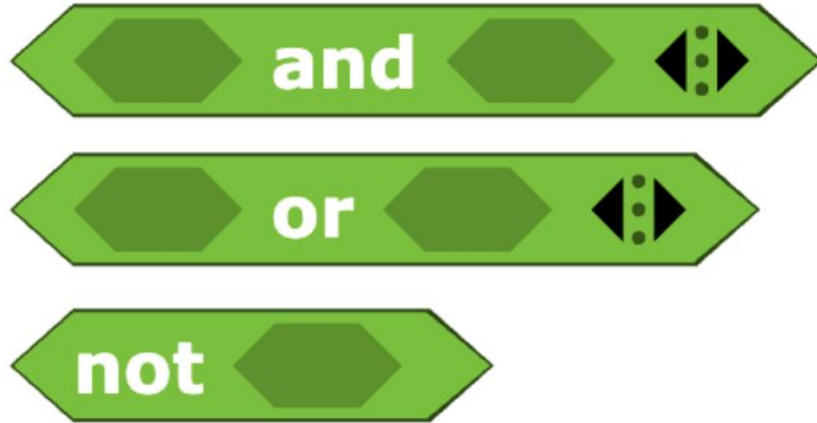
The only operations that can be performed at the lowest level are those that operate on **single-bit values**

0 or 1

on or off



Boolean (or Logical) Function



Notice that both the blocks themselves and the input slots in the blocks are hexagonal.

Boolean functions take Boolean values (True or False) as **inputs** and report a new Boolean value as **output**.



Boolean → Data Representation

OR Blocks | Evaluates to

 False

 True

 True

 True

OR evaluates to **true**, as long as **one operand evaluates to true**



Boolean → Data Representation

OR Blocks | Evaluates to

 False

 True

 True

 True

AND Blocks | Evaluates to

 False

 False

 False

 True

OR evaluates to **true**, as long as **one operand evaluates to true**
AND evaluates to **true**, only if **all operands are true**



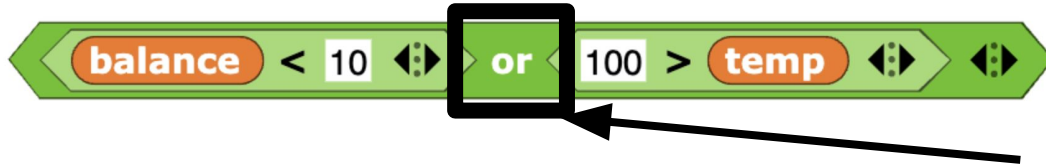
Boolean Function Examples



balance and temp are variables



if balance is less than 10 **AND**
if temp is greater than 100



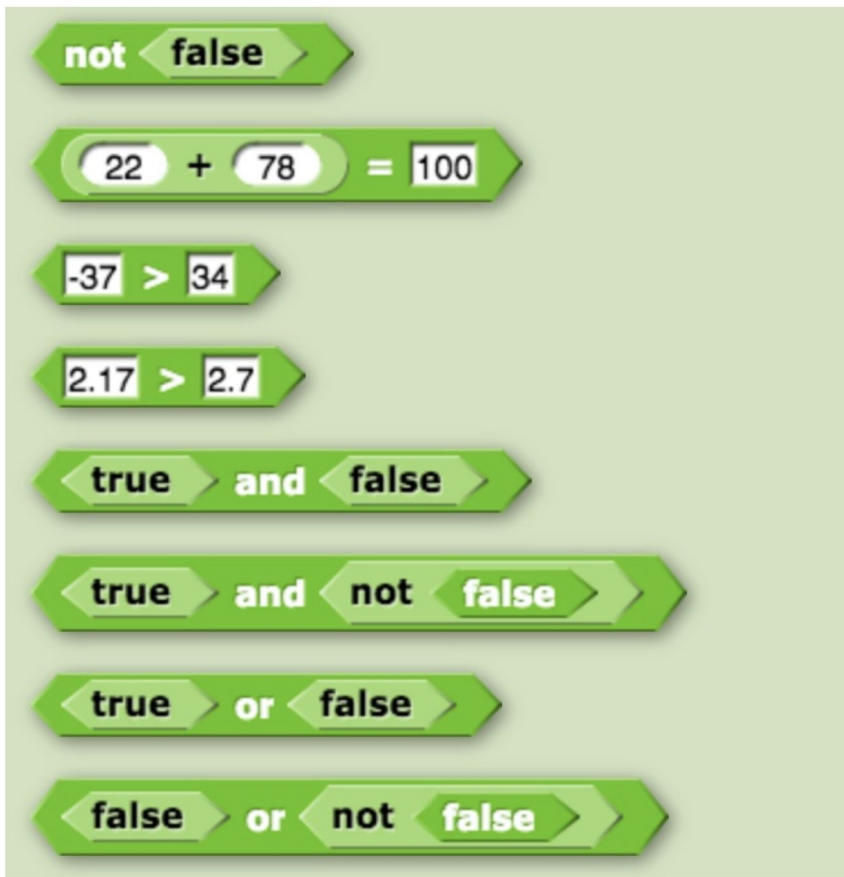
if balance is less than 10 **OR**
if temp is greater than 100



if balance is **NOT** less than 10

Class Activity

What does each block evaluate to?



A collection of Scratch-style logic blocks arranged vertically on a light green background. The blocks are as follows:

- not** block with a **false** input.
- =** block with inputs **22**, **78**, and **100**.
- >** block with inputs **-37** and **34**.
- >** block with inputs **2.17** and **2.7**.
- and** block with inputs **true** and **false**.
- and** block with inputs **true** and **not false**.
- or** block with inputs **true** and **false**.
- or** block with inputs **false** and **not false**.

True?

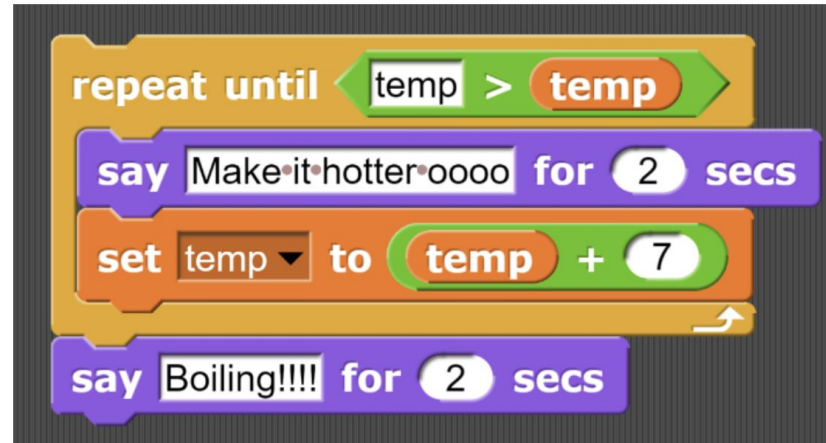
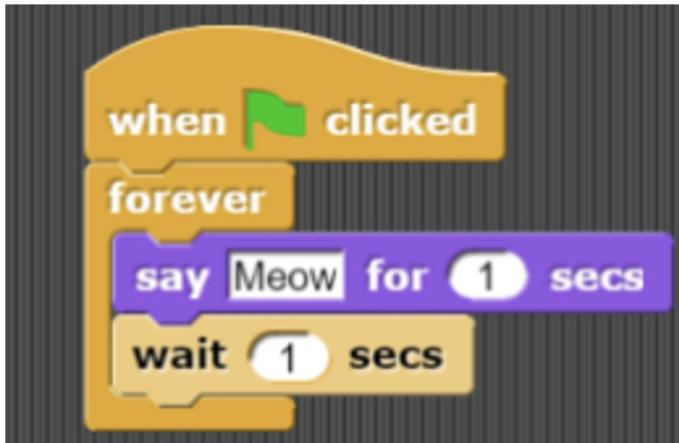
OR

False?

Iteration

What if you want to do a task over and over again?

A **loop** allows you to do the same task over & over again, sometimes with a **stopping** condition, sometimes **forever**!



Repeat Blocks

Repeat some code a finite number of time



Repeat UNTIL a particular condition has been met.

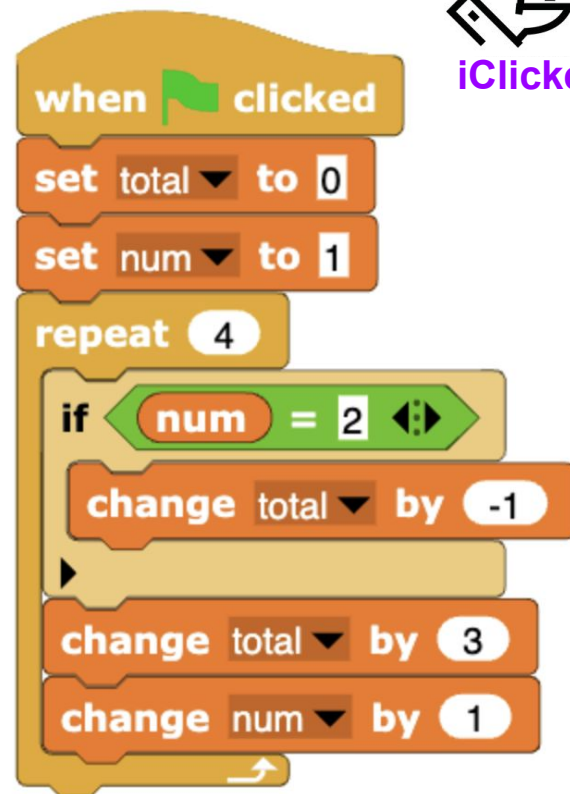
If the condition is never met, then, it goes on *forever*.



Q: What is the value in **total** and **num** when this code block is run?



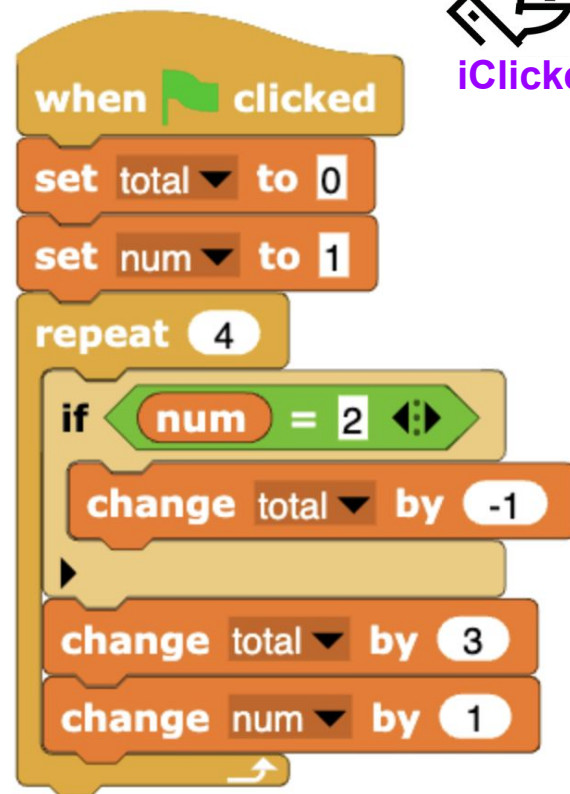
Iteration	num (Before)	Condition Check (num = 2?)	Change to total (-1 if true)	total +3	total (After)	num +1	num (After)
1st	1	No (skip -1)	0	+3	3		
2nd							
3rd							
4th							



Q: What is the value in **total** and **num** when this code block is run?



Iteration	num (Before)	Condition Check (num = 2?)	Change to total (-1 if true)	total +3	total (After)	num +1	num (After)
1st	1	No (skip -1)	0	+3	3	+1	2
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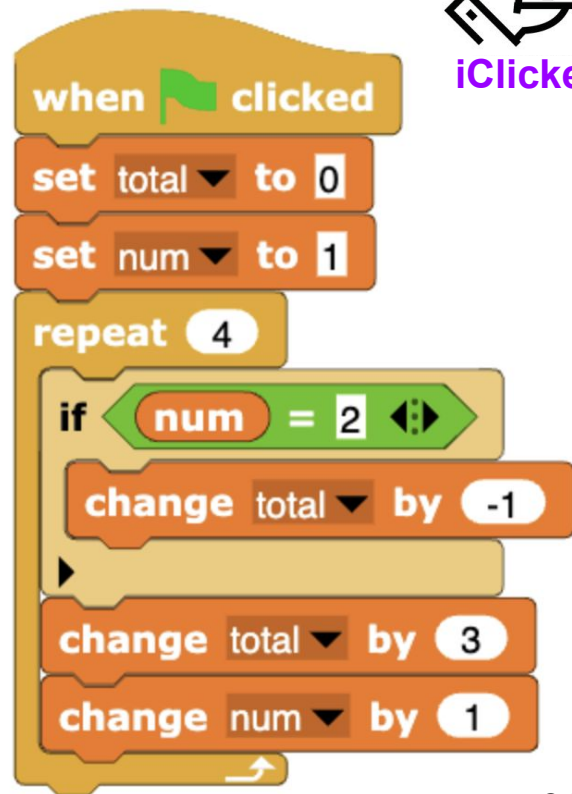


Q: What is the value in **total** and **num** when this code block is run?



iClicker

Iteration	num (Before)	Condition Check (num = 2?)	Change to total (-1 if true)	total +3	total (After)	num +1	num (After)
1st	1	No (skip -1)	0	+3	3	+1	2
2nd	2	Yes (-1 applied)	$3 - 1 = 2$	+3	5	+1	3
3rd	3	No (skip -1)	5	+3	8	+1	4
4th	4	No (skip -1)	8	+3	11	+1	5



Q: What is the value in *i* when the code is run, assuming user input = 3?

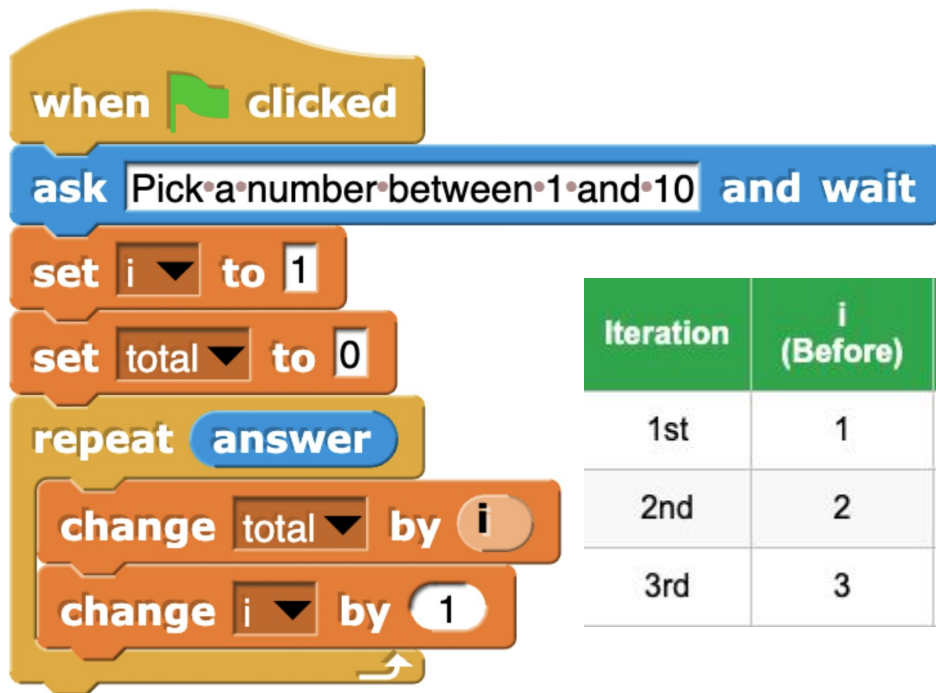


- A. 1
- B. 2
- C. 3
- D. 4
- E. 5

```

when clicked
  ask Pick a number between 1 and 10 and wait
  set i to 1
  set total to 0
  repeat answer
    change total by i
    change i by 1
  
```

Q: What is the value in **i** when the code is run, assuming **user input = 3**?



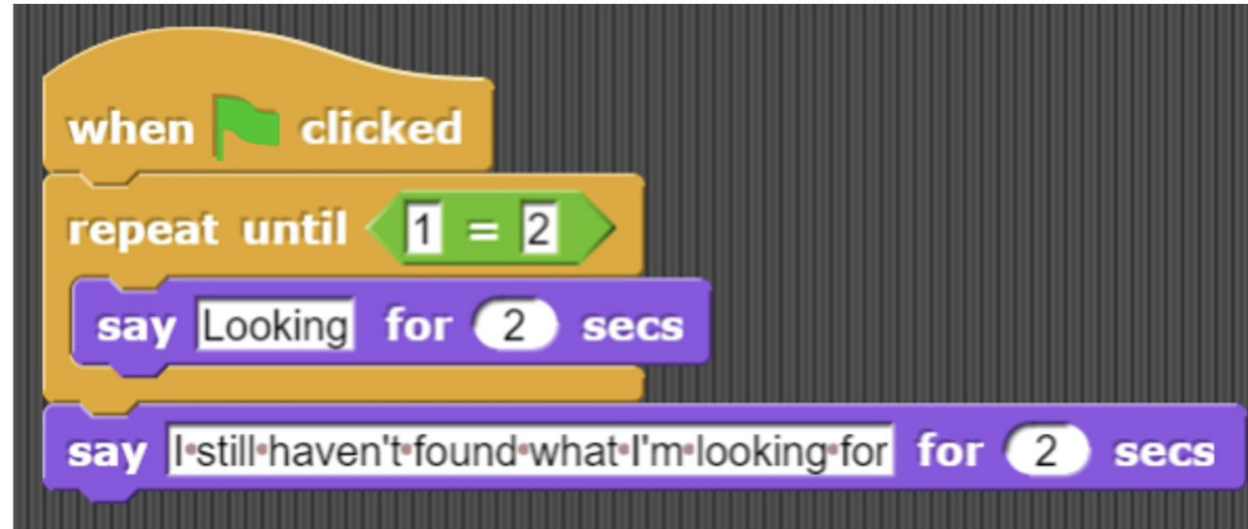
Iteration	i (Before)	total (Before)	Add i to total	total (After)	i +1	i (After)
1st	1	0	$0 + 1$	1	+1	2
2nd	2	1	$1 + 2$	3	+1	3
3rd	3	3	$3 + 3$	6	+1	4



Q: Will this program ever say “I still haven’t found what I’m looking for”?



- A. Yes
- B. No
- C. Sometimes



For loops

when  clicked

for **i** = 1 to 10

say **i** for 1 secs

say Bye-bye! for 2 secs

when  clicked

for  = 1 to 10

i is a variable

say  for 1 secs

say Bye-bye! for 2 secs

when  clicked

for  = 1 to 10

say  for 1 secs

say Bye-bye! for 2 secs

**i is initialized
with a value of 1**

when  clicked

for **i** = 1 to 10

say **i** for 1 secs

say Bye-bye! for 2 secs

This loop will run until i has a value that is not between 1 to 10 (inclusive)

when  clicked

for **i** = 1 to 10

say **i** for 1 secs

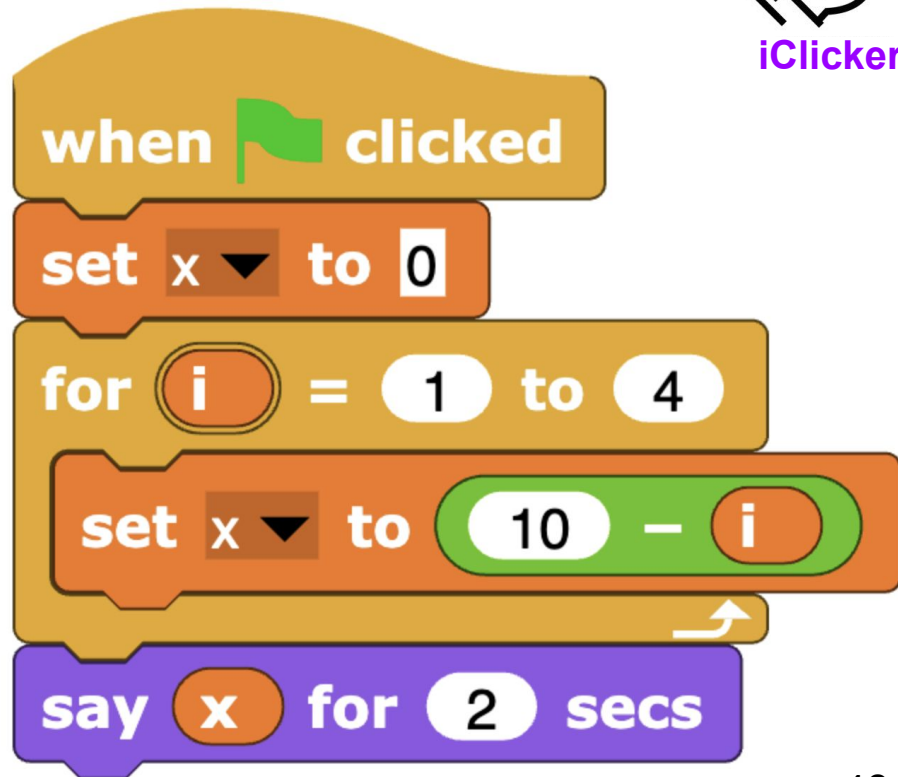
say Bye-bye! for 2 secs

Every time we reach the end of the loop, i will increase (increment) by 1

Q: What is the value in **x** when the code is run?



- A. 1
- B. 3
- C. 4
- D. 6
- E. 10



Loops

Repeat Blocks

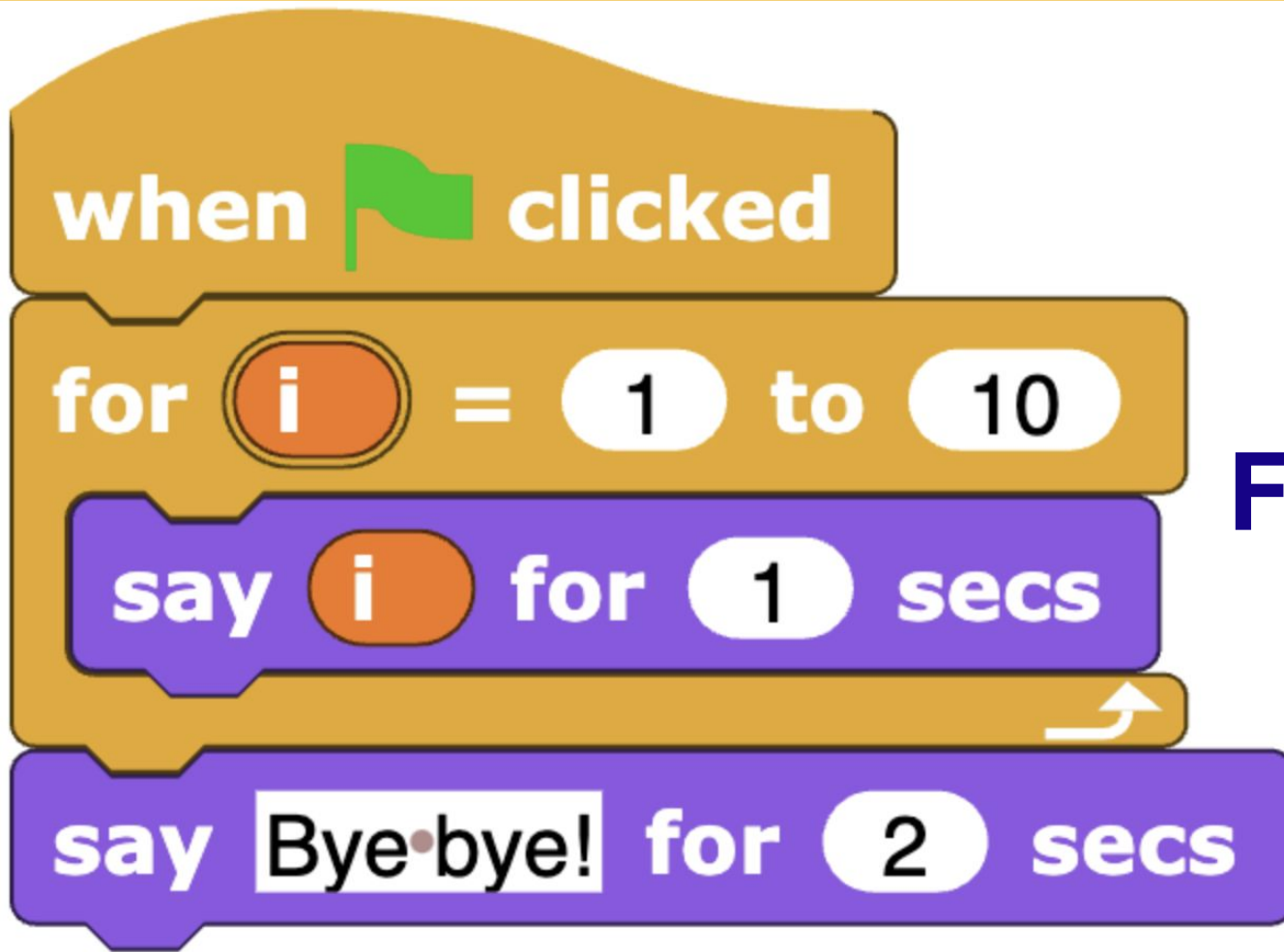
Repeat some code a finite number of time



Repeat UNTIL a particular condition has been met.

If the condition is never met, then, it goes on *forever*.

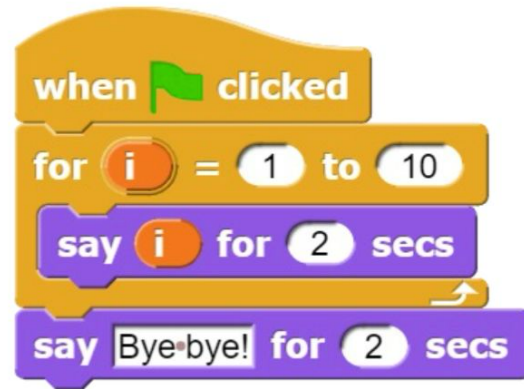
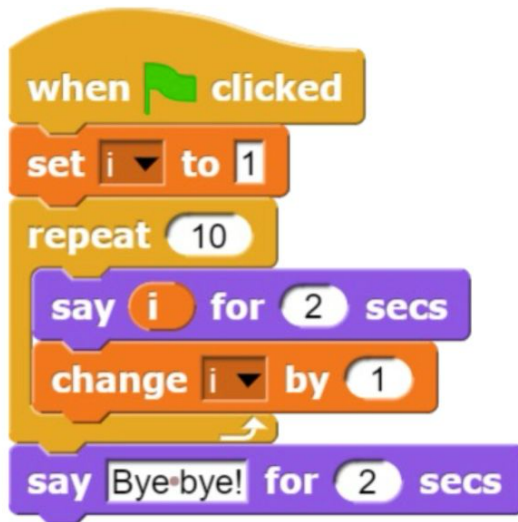
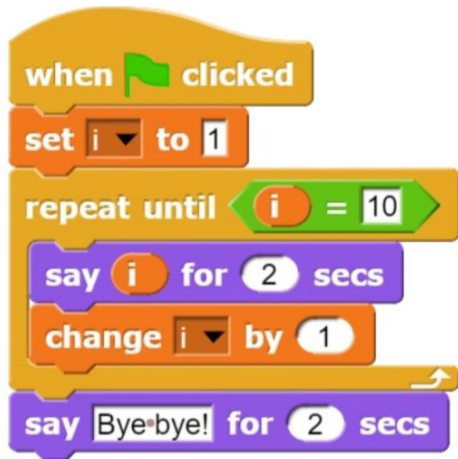




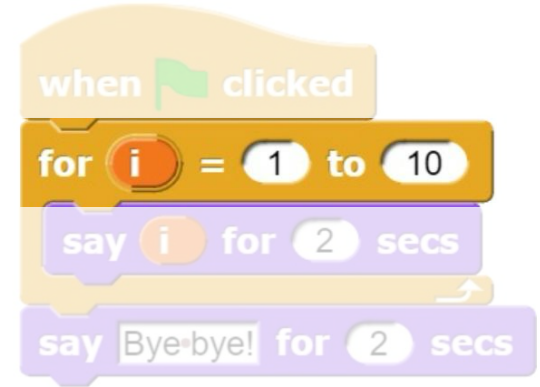
For Loops

Comparing Loops

What's the difference between these loops?



What's the difference between these loops?



Wrap up